Data-driven recommendations for increasing reliability in measures of infant word recognition

Veronica Boyce[i], Virginia A. Marchman[a], Heidi Baumgartner[a], Claire Augusta Bergey[b], Mika Braginsky[a], Federico Giovannetti[c], George Kachergis[a], Kyle MacDonald[j], Jess Mankewitz[d], Stephan Meylan[e], Ben Prystawski[a], Robert Z. Sparks[a], Adrian Steffan[f], Alvin Wei Ming Tan[a], Tian Linger Xu[g], Michael C. Frank[a], & Martin Zettersten[h]

[a] Department of Psychology, Stanford University, Stanford, CA, USA
[b] Department of Linguistics, Stanford University, Stanford, CA, USA
[c] Unidad de Neurobiología Aplicada (UNA, CEMIC-CONICET), Buenos Aires, Argentina
[d] Department of Psychology, University of Wisconsin, Madison, WI, USA
[e] Department of Linguistics, University of California, Berkeley, CA, USA
[f] Department of Psychology, Ludwig Maximilian University, Munich, Germany
[g] Department of Psychology, Indiana University, Bloomington, IA, USA
[h] Department of Cognitive Science, University of California, San Diego, CA, USA
[i] Department of Brain and Cognitive Sciences, Massachusetts Institute of Technology, Cambridge, MA, USA
[j] IBM

Author Note

AI Disclosure statement: Generative AI was used to do code refactoring and code review.

Abstract

Children's early word recognition is both an important phenomenon in its own right and a key index of early language more broadly. A key paradigm for measuring word recognition in young children is the looking-while-listening paradigm (LWL), which tracks eye movements to the referent of a word as a measure of online language comprehension. Yet the analytic and design choices for LWL vary widely, and the consequences of these choices have rarely been investigated at a large scale. Here we use data from Peekbank, a repository of LWL data from studies with almost 4000 children (6 months – 5 years), to search for analytic and design decisions that maximize the reliability and validity of LWL measures. Analytic recommendations include using long analytic windows for computing accuracy and avoiding baseline correction. We also find little or no benefits from excluding children or trials with low data. The recommended combination of analytic choices increased reliability by approximately .20 and validity by .07. Design recommendations include building experiments with a minimum of 10 target trials per child per condition to improve reliability and matching stimuli for animacy because image saliency differences can mask word knowledge in infants. We hope that these recommendations lead to improved and unified standards for the measurement of children's word recognition in the LWL paradigm.

Keywords: looking while listening, word recognition, language development, eyetracking, reliability, validity

Word count: X

Data-driven recommendations for increasing reliability in measures of infant word recognition

Introduction

Over the first few months of life, children become experts in their native language, organizing the phonological system of speech input and parsing speech into meaningful units. Soon after, children begin to show early signs of comprehension of words or phrases through simple behaviors such as turning their head when their name is called (e.g., “Hey, Billy!”) or reaching for an object that is named (e.g., “Where’s the ball?”). These everyday accomplishments mark important milestones in children’s development. Yet global assessments of these milestones do not capture the millisecond-level processes that take place from moment-to-moment during real-time language comprehension. Children, like adults, process speech incrementally, as it unfolds in time (Fernald, Pinto, Swingley, Weinberg, & McRoberts, 1998; Frank et al., 2025; Snedeker, 2013), and use features of the speech signal to extract meaning that supports language comprehension (Law & Edwards, 2015; Weisleder & Fernald, 2013).

These moment-to-moment processes are difficult to document without child-friendly procedures that precisely measure how children’s eye-movements are time-locked to speech. In particular, the “looking while listening” (LWL) procedure has emerged as a *de facto* standard for assessing language comprehension in infants and children, especially when they are younger than two years old, as other options for direct assessment during this period are limited (Frank, Braginsky, Yurovsky, & Marchman, 2021).[1] In the standard version of LWL, children see two pictures on a screen and hear speech directing their attention to one of the pictures, e.g., “look at the ball”. The accuracy and speed of fixating the named referent, rather than the distractor image, measure the efficiency of language comprehension (Fernald et al., 2008).

LWL has been a key tool for enhancing our understanding of the continuous processes of growth in incremental language processing (Golinkoff, Ma, Song, & Hirsh-Pasek, 2013). For example, using LWL methods, we can measure developmental increases in the speed and accuracy of children’s recognition of familiar words (Fernald, Perfors, & Marchman, 2006; Frank et al., 2025) and in their sensitivity to detailed phonological information during the first years after birth (Moore & Bergelson, 2024; Swingley & Aslin, 2000). Moreover, measures of children’s skill in the LWL task have proven useful as an index of individual differences. Children who show more efficient language processing of familiar words at 18 months show larger vocabulary size and more rapid vocabulary growth over the second and third years of life than children with less efficient processing (Fernald & Marchman, 2012; Fernald et al., 2006; Frank et al., 2025; Peter et al., 2019). Variation in language processing efficiency has also been shown to be associated with later language, cognitive, and non-verbal outcomes in both typically-developing (Law & Edwards, 2015; Peter et al., 2019) and preterm children (Loi, Marchman, Fernald, & Feldman, 2017; Marchman, Adams, Loi,

[1] Here we adopt LWL as a convenient label while recognizing that the procedure goes by a number of different names in the literature including the “intermodal preferential looking procedure” (Fernald, Zangl, Portillo, & Marchman, 2008; Hirsh-Pasek & Golinkoff, 1996; Reznick, 1990).

Fernald, & Feldman, 2016; Marchman et al., 2018).

One key reason for the success of this research is the care that has been invested in the development of the LWL method. Past work has articulated best practices in collecting and analyzing LWL measures, with a particular focus on stimulus and design choices that are assumed to ensure that looking preferences are maximally informative about word recognition processes (Fernald et al., 2008). However, despite widespread use of the LWL method in language development research, there is still limited consensus on how to best process and analyze time-series data of this type (from either adults or infants, Mirman, Dixon, & Magnuson, 2008; Bergelson & Swingley, 2012; Peelle & Van Engen, 2021; Von Holzen & Mani, 2012; Weaver, Saffran, & Arunachalam, 2026).

In the developmental field especially, this lack of consensus likely stems from structural factors: infant recruitment is time- and cost-intensive and datasets collected by individual labs are often small (Bergmann et al., 2018; Frank, Braginsky, Yurovsky, & Marchman, 2017). These issues make it difficult for any one researcher to systematically evaluate the consequences of specific analytic choices independently of the data collected for a particular study (cf. Peelle & Van Engen, 2021). Instead, researchers are often forced to make ad-hoc decisions or to adopt idiosyncratic lab practices that themselves originate from subjective judgments of what "seems to work". By decreasing uniformity in the literature, this variation also increases the challenge of triangulating what precisely is being measured by LWL (Weaver et al., 2026). How can we put the measurement and analysis of infant word recognition and online language processing on firmer empirical footing?

Peekbank is a new data resource that offers a potential solution to this core issue by providing a large database of infant eye-tracking data from the LWL paradigm for exploring methodological questions (Zettersten et al., 2022). The most recent release of Peekbank (version 2026.1) houses 44 datasets from infants across a wide age range (6 months – 5 years) and across a broad set of test words and materials, resulting in observations from over 3915 infants participating in the LWL task. The goal of the current manuscript is to use these data to systematically explore key analytic decisions that researchers make when using this paradigm, offering best-practices guidance for measurement.

Our approach throughout is to seek analytic choices that distill the best possible measures from the rich time course of behavior in the LWL task. Recent reviews of the LWL literature highlight open questions about both the reliability and validity of looking-based measures of word recognition (Weaver et al., 2026). Starting with reliability, we aim for analyses that yield the maximal signal about individual children relative to both measurement error and irrelevant artifacts (Byers-Heinlein, Bergmann, & Savalei, 2022; Rhemtulla & Bottesini, 2022; Schreiner et al., 2024). Here, to measure reliability, we compute both within-dataset inter-item reliability and test-retest reliability for datasets with multiple LWL sessions close in time for the same children.

Turning to validity, the goal is to make analytic decisions that maximize the relationship with other measures that are thought to reflect related constructs (Flake &

Fried, 2020; Kominsky, Lucca, Thomas, Frank, & Hamlin, 2022). Here, we select as our primary measure the correlations between LWL measures and parent reports of vocabulary measured with the MacArthur-Bates Communicative Development Inventories (MB-CDI), a popular parent report measure of early vocabulary (Frank et al., 2021; Marchman, Dale, & Fenson, 2023) that is widely used in conjunction with LWL tasks (CITE Fernald etc). We focus specifically on the relationships between aggregate measures of receptive and expressive vocabulary size from the MB-CDI and LWL task, given the documented relationships between these measures (e.g., Fernald & Marchman, 2012; Peter et al., 2019). This correspondence reflects the general relationship between familiar word recognition in the moment (indexed by LWL) and either concurrent vocabulary size or the growth of early vocabulary over time (indexed by the MB-CDI) (Frank et al., 2025). We do not explore relations at the item-level (e.g., speed of processing of a particular word and a child's reported knowledge of that word) given the mixed evidence for item-level alignment between LWL and MB-CDI data (López Pérez, Moore, Sander-Montant, & Byers-Heinlein, 2024; Weaver & Saffran, 2026).

Our focus in this paper is on providing guidance for navigating some of the many decisions facing an investigator analyzing LWL data. Recent work using an earlier version of Peekbank (as well as other datasets) has investigated the psychometric properties of a range of different dependent measures in LWL tasks and their general construct structure (Sander-Montant, Fibla, Byers-Heinlein, & Killam, 2025). We focus instead on the consequences of varying analytic choices for two main measures, which tend to have the strongest psychometric properties (Sander-Montant et al., 2025). The first is *accuracy*: how much of the time a child looks at the target image compared to looking at either the target or distractor image over some period of time after the auditory stimulus. Children's responses unfold over time, but it is not known what start and end time maximizes the signal of children's response to the stimulus (Peelle & Van Engen, 2021). The second measure we focus on is *reaction time* (RT): how long it takes a child to change where they are looking in response to the stimulus (Fernald et al., 2008). There are even more analytic options available for this metric — whether to measure RT as when a child shifts attention away from the distractor (launch-based RT) or when a child completes this shift and fixates the target (land-based RT), whether to measure RT on a log or raw scale, and how long after the start of the target onset to start counting the RTs. For all measures, there are also questions about whether and when to exclude either children or trials.

While the LWL task is used for a wide variety of purposes, including measurements of individual differences (e.g., Peter et al., 2019) as well as measurements of experimentally-induced condition differences (e.g., Swingley & Aslin, 2000), here we focus only on what we refer to as "vanilla" word recognition items. These are items that test vocabulary directly with familiar targets, familiar distractors, and no informative carrier phrases or experimental manipulations. We focus on these items both because they are the most commonly used LWL items and because they are measurements in which reliability and validity are most clearly defined. Vanilla trials are expected to reflect the construct of word knowledge and correlate with other measures of word knowledge, including vocabulary size.

The plan of the paper is as follows. We begin by overviewing the subset of the Peekbank database we selected for our analyses. We then make a set of targeted recommendations, providing evidence for each based on the impact of our recommendation on the average reliability and validity of the resulting measures. Our first set of recommendations focus on data processing decisions and our second set of recommendations concern design considerations. To foreshadow our results, we find that while many common data analysis practices provide relatively reasonable levels of reliability and validity, it may be possible to further optimize reliability and validity through careful, data-driven decision-making.

Methods: Dataset and approach

Peekbank is an aggregation of datasets from a number of LWL studies. This work is based on Peekbank version 2026.1 (released February 2026) which has 134929 trials from 3915 children and 44 datasets.

Inclusion criteria

We limit our analyses to what we term "vanilla" trials: trials that represent standard LWL trials testing familiar words without additional experimental manipulations. For a trial to be considered "vanilla", both the target and the distractor images must be of familiar objects. For the auditory stimulus, the target word must be a real word that occurs once and is the first and only source of information about the target. The carrier phrase must be a well-formed, grammatical phrase, without any switches in language, and without any intentionally added noise or audio filtering. Vanilla trials could come from studies with only vanilla trials, from the filler or baseline trials of experimental studies, or even from studies where the experimental conditions met the vanilla criteria. (See SI: Criteria for Inclusion for more detailed criteria used to handle borderline situations.) While still heterogeneous, these "vanilla" trials have a shared interpretation for how looking behavior is expected to relate to underlying language knowledge and processing. We call the start of the target word in the audio the stimulus onset and label this as the zero point for timing. We only include trials from children 60 months or younger (see Figure S1 for age distributions). We do not place restrictions on the language of the trials or children's language background.

Descriptives

This paper is based on 36 datasets, comprising 3171 children, 4851 administrations, and 75636 trials. There are MB-CDI comprehension scores for 632 children from 12 of the datasets, and MB-CDI production scores for 1143 children from 20 of the datasets. Our test-retest data comes from 5 datasets with multiple administrations within 1.5 months of each other, covering 365 children (788 administration pairs). See Table 1 for a list of included datasets. Figure S2 has time course plots showing the number of trials and timing of available data for each dataset.

Table 1
Descriptives of the included datasets.

	Dataset Name	Pct Trials	Trials	Admins	Subjects	Age	CDI	Test-retest	Language
1	Adams et al. (2018)	18.4%	13899	711	69	22 [13 - 38]	x	x	English
2	Fernald & Marchman (2012)	15.6%	11828	679	122	22 [17 - 32]	x	x	English
3	Weaver et al. (2024)	6.5%	4948	248	141	16 [14 - 24]		x	English
4	Fernald et al. (2013)	5.8%	4359	178	80	20 [17 - 26]	x	x	English
5	Hurtado et al. (2008)	5.3%	4035	136	76	21 [17 - 27]	x		Spanish
6	Fernald et al. (2006)	4.9%	3692	229	63	20 [15 - 25]	x		English
7	Yurovsky et al. (2013)	3.7%	2810	413	413	34 [12 - 60]			English
8	Yurovsky et al. (2017)	3.1%	2350	327	327	37 [8 - 60]			English
9	Bergelson & Swingley (2012)	2.8%	2081	84	84	12 [6 - 21]	x		English
10	Kartushina & Mayor (2019)	2.7%	2072	69	69	7 [6 - 9]	x		Norwegian
11	Weaver & Saffran (2026)	2.4%	1788	69	69	19 [18 - 20]	x		English
12	Weisleder & Fernald (2013)	2.3%	1769	58	29	22 [18 - 27]			Spanish
13	Yurovsky & Frank (2017)	2.3%	1747	285	285	26 [13 - 59]			English
14	Borovsky & Peters (2019)	2.2%	1676	79	79	18 [17 - 20]	x		English
15	Potter & Lew Williams (2023)	2.1%	1626	67	67	24 [21 - 27]	x		English
16	Yoon et al. (2015)	2.0%	1500	199	199	43 [13 - 60]			English
17	Sander-Montant et al. (2023)	1.8%	1381	123	123	22 [12 - 31]	x		English, French
18	Mahr et al. (2015)	1.6%	1232	29	29	21 [18 - 24]	x		English
19	Egger et al. (2020)	1.4%	1052	54	54	18 [18 - 19]	x		Dutch
20	Garrison et al. (2020)	1.4%	1046	35	35	15 [12 - 18]	x		English
21	Hurtado et al. (2007)	1.3%	1010	49	49	24 [15 - 37]			Spanish
22	Pomper & Saffran (2017)	1.0%	785	82	82	17 [14 - 19]			English
23	Frank et al. (2016)	1.0%	781	108	108	35 [12 - 60]			English
24	Bacon & Saffran (2022)	1.0%	754	38	38	23 [22 - 24]	x		English
25	Ronfard et al. (2022)	1.0%	737	40	40	20 [18 - 24]	x		English
26	Perry & Saffan (2017)	0.9%	718	42	42	20 [19 - 22]	x		English
27	Casillas et al. (2017)	0.9%	651	23	23	32 [9 - 48]			Tseltal
28	Swingley & Aslin (2002)	0.8%	596	50	50	15 [14 - 16]	x		English
29	Pomper & Saffran (2016)	0.6%	471	60	60	44 [41 - 47]			English
30	Kremin et al. (2023)	0.5%	410	60	60	42 [37 - 49]			English, French, Spanish
31	Pomper & Saffran (2015)	0.5%	389	25	25	40 [38 - 42]			English
32	Moore & Bergelson (2022)	0.5%	384	29	29	18 [16 - 20]	x		English
33	Potter et al. (2019)	0.4%	309	44	23	23 [18 - 29]	x	x	English
34	Byers-Heinlein et al. (2017)	0.4%	284	48	48	20 [19 - 21]	x		English, French
35	Pomper & Saffran (2019)	0.3%	249	44	44	40 [38 - 43]			English
36	Pomper & Saffran (2018)	0.3%	217	37	37	39 [38 - 43]			English
	Total	100%	75636	4851	3171	22 [6 - 60]	21	5	

Approach

Throughout we use two reliability metrics and two validity metrics. Reliability refers to how consistently measurements are measuring the same thing. One of our measures of reliability is an intra-class correlation (ICC). Specifically, we use a measure of inter-rater reliability that measures (within a dataset) how much different items (target labels) are consistent with each other in "rating" the subjects (kid-administrations). See SI: Details about ICC for more specifics. Another measure of reliability is test-retest reliability, which measures the correlation for measures on the same child at two different administrations within 1.5 months of each other. This measure is only possible on a subset of the datasets that closely spaced repeat administrations. While we are limited in how much data we have for test-retest reliability, this is the metric that is most important for measuring stable individual differences.

In addition to having reliable measures, we also care that our measurements correspond to the construct of interest (vocabulary and language processing speed). For validity, we correlate with CDI production measures and CDI comprehension measures, from the datasets where we have CDI administrations. We use both short and long form CDIs, using the proportion of items a child was marked as knowing as our measure. In general, correlating with production and comprehension in the expected directions is desirable, although we also may expect that LWL measures are detecting distinct aspects of language learning compared to what is represented in the CDI (see Frank et al., 2025 for a detailed analysis).

All the code and data to reproduce the analyses is available at https://github.com/peekbank/peekbank-method [TODO: anonymize for review if needed].

Data processing recommendations

Our first set of recommendations focuses on data processing decisions. LWL data undergoes significant post-processing.. Experiment design, eye-tracking, and eye-gaze coding considerations are covered elsewhere (Fernald et al., 2008; Olson et al., 2020; Venker et al., 2020; Wass, Smith, & Johnson, 2013), so we focus here on the process of going from a series of timepoints to the measures of accuracy and reaction time that are analysed. There are a number of decisions involved, including what windows and time cutoffs to use, and what exclusions to make around missing or low amounts of data.

Recommendation 1: Measure accuracy over a long window.

In LWL studies, accuracy is typically calculated as the fraction of time a child is looking at the target divided by the time they are looking at the target or the distractor, over a time window of interest. A key decision is therefore the time window of analysis. Typically windows start shortly after the point of disambiguation and continue until a few seconds later; this window is intended to capture the peak of children's stimulus-driven looks to the target. However, in practice, studies vary in what windows are used, and often window selection is a locus for analytic flexibility (choosing windows post hoc to maximize a particular effect) (Peelle & Van Engen, 2021). Common analysis windows include starting 300 milliseconds after stimulus onset and ending 1800 ms after stimulus onset (following Fernald et al., 2008) and 367ms to 2000ms after stimulus onset (e.g., Swingley & Aslin, 2002), although windows that extend until 3 or 4 seconds after stimulus onset are also used (e.g., Bergelson & Swingley, 2012; Bergelson & Aslin, 2017; López Pérez et al., 2024). In this analysis, we begin from first principles and ask what windows maximize reliability and validity on word recognition trials.

Analytic approach. We first calculated the ICC reliability for accuracy across datasets, computing average ICC across a range of different window onsets and offsets. Next, we added the additional metrics of test-retest reliability and—to assess validity—CDI correlations, which were calculated by dataset for each combination of parameter values. To aggregate across datasets, we used a linear mixed-effects model with random intercepts by dataset, weighted by the number of administrations. We interpret the fixed-effect contrasts as what differences in outcome measure would be expected from switching from one set of parameters to another on a generic dataset. We report estimates and 95% confidence intervals. The best data-processing options might be different for children of different ages, so we ran the same linear mixed effect models separately for data just on children in each of 4 age groups: <18 months, 18-24 months, 24-36 months, and >36 months.

Results. Average ICCs across windows are shown in Figure 1. As expected, including time points from before the point of disambiguation decreases reliability. Our

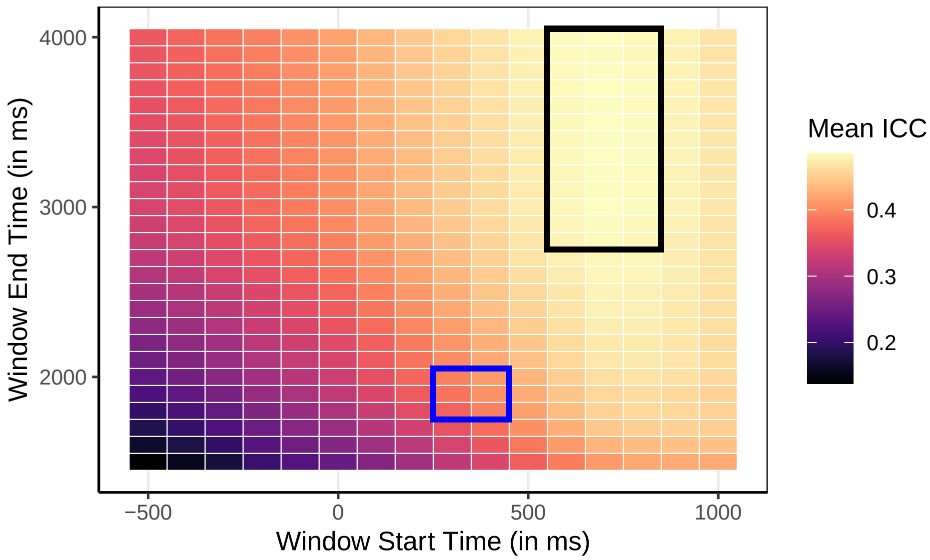

Figure 1. A heat map showing the average ICC across datasets for different start and end points for the accuracy window. A region with the highest ICC (best reliability) is outlined in black, and the region most commonly represented in the literature is outlined in blue.

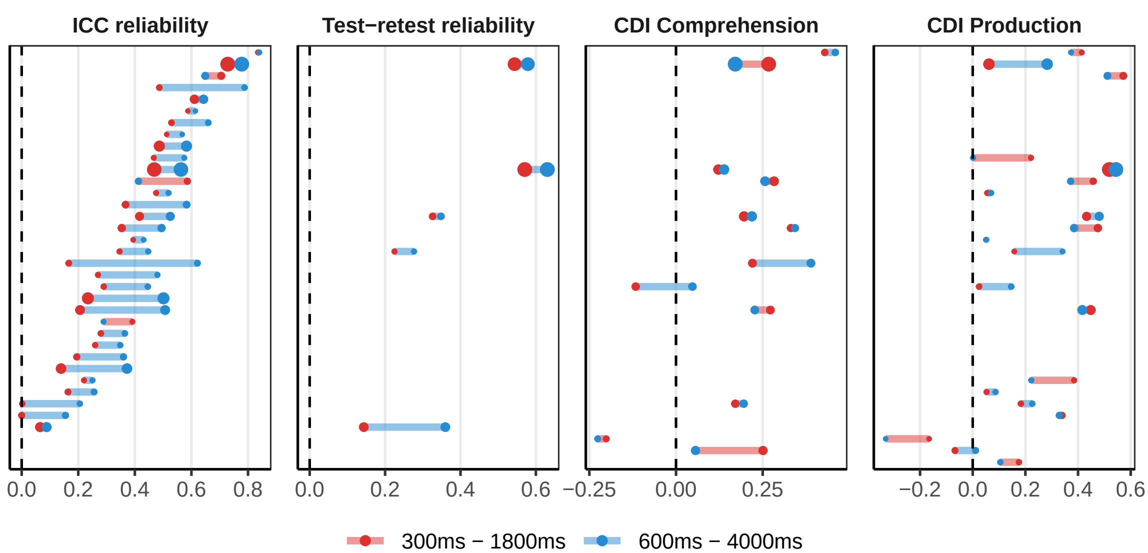

Figure 2. Per dataset outcome measures for two window options: 300ms-1800ms (red) and 600ms-4000ms (blue). Dot size is proportional within each panel to the number of administrations.

initial exploration found a region of high reliability with onsets around 600-800ms and offsets around 2800-4000ms. A dataset by dataset comparison of the differences between a 300-1800ms window (representative of common practice) and a 600-4000 ms window are shown in Figure 2. Nearly all datasets have higher ICC reliability for the later and longer window. All 5 test-retest reliability datasets with close in time pairs of administrations also show increased reliability on this later and longer window. Our mixed effect model confirms that there are expected reliability gains from switching from a 300-1800ms window to a

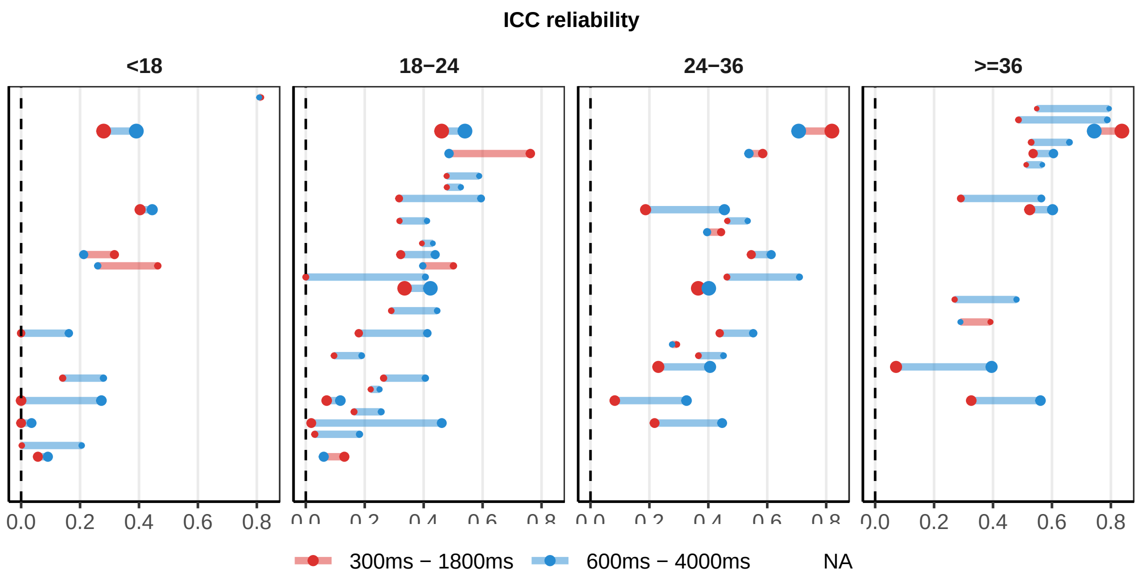

Figure 3. Per dataset ICC measures for two window options: 300ms-1800ms (red) and 600ms-4000ms (blue), shown split by age category. Dot size is proportional within each panel to the number of administrations.

600-4000ms one. ICC reliability is expected to increase by 0.12 [0.08, 0.16] and test-retest reliability by 0.07 [-0.02, 0.16]. There is no evidence of differences in validity between these two windows (-0.02 [-0.07, 0.03] for vocabulary comprehension or 0.03 [-0.02, 0.08] for vocabulary production).

These patterns generally hold across age. The overall level of reliability rises with age, as older children generally fixate target images more consistently and experiments with younger children typically use fewer trials (Figure S3). However, the overall pattern of which time windows maximize reliability and validity at any given age is remarkably consistent across the different age groups (Figure S5). For instance, across different ages there is an improvement in ICC for using a 600-4000 window compared to a 300-1800ms one (Figure 3). For <18 months, the ICC difference is 0.06 [-0.03, 0.16]; for 18-24 months, 0.13 [0.07, 0.19]; for 24-36 months, 0.06 [0.01, 0.12]; and for 36+ months 0.15 [0.06, 0.23].

While we focused our analyses on the window that maximized reliability numerically (600 - 4000 ms), there are a wide range of window options with generally comparable reliability and validity measures (Figure S4). For instance, a 400 - 3000 ms window is not significantly different from a 600 - 4000 ms window on any of the outcome measures. Although we recommend starting between 400ms and 600 milliseconds to optimally balance validity and reliability, choosing when precisely the accuracy window starts is likely less important for maximizing reliability than choosing a long time window: regardless of window start time, increasing the window size consistently improves reliability (Figure 1).

Discussion. Overall, we find that LWL accuracy measures are reliable across a wide range of window choices; however, to maximize reliability, we recommend windows that start somewhere in the 400-600 ms range and extend to 3000-4000 ms (longer than the more canonical window ends of 1800 or 2000 ms). Windowing choices seem to primarily affect reliability measures and have limited effects on our validity measures (receptive and productive vocabulary). Why do later starting and later ending windows increase reliability? This suggests that, even as looking to the target often begins to decline in the post 2000ms window, looking behavior still contains reliable signal; at the same time, there may relatively little reliable signal in the early 300-600ms looking period. While some children may look quickly to the target and then look away, other children who look to the target later or who continue looking at the target at still contributing signal during later portions of a given trial. Overall, children in many datasets are in aggregate looking to the target more than the distractor during later looking periods (even if less than at the peak, Figure S2).

Recommendation 2: Do not use baseline-corrected accuracy.

One commonly acknowledged issue in LWL studies is that looking behavior could be partly driven by differences in visual salience between the target and distractor images themselves (Fernald et al., 2008; Novack et al., 2025): Children may have a baseline preference to look at one image over another, even in the absence of auditory stimuli. Although thoughtful design choices can minimize these issues (see Recommendation 6), some analytic strategies also attempt to account for baseline preferences in dependent measures. While there are a variety of such strategies in the LWL literature (e.g., computing difference scores for specific target-distractor pairs; controlling for baseline preference), here we focus on one common practice: "baseline correction". In baseline correction, researchers use a difference measure between accuracy on a post-stimulus window and "accuracy" (looks to target versus looks to target or distractor) on a pre-stimulus window (e.g., Bergelson & Swingley, 2012; Weaver, Zettersten, & Saffran, 2024). The pre-stimulus window can be the 2 seconds prior to disambiguation (Weaver et al., 2024) or the full 3-4 second pre-stimulus period (Bergelson & Swingley, 2012).

Analytic approach. We considered baseline-corrected accuracies with various pre-stimulus windows, focusing on a a 600ms-4000ms main window (although the results generalize to other main windows). We restricted ourselves to datasets with at least some data from 2000ms or more before stimulus onset (see Figure S2 for which datasets have data when). We also computed a "no baseline" accuracy, based solely on accuracy within a 600-4000ms time window with no correction. Baseline windows started at -1000, -2000, -3000 and -4000ms and ended at stimulus onset.

Results. Baseline-corrected accuracy led to worse reliability and validity than accuracy. As an illustrative example, Figure 4 shows per-dataset results for accuracy compared to a baseline-corrected accuracy using a -2000 - 0 baseline window. This result held generally: All baseline window sizes that we considered led to worse reliability and validity measures compared to using normal accuracy without any correction. For a regular window of 600-4000, baseline correction is expected to decrease ICC by at least 0.1 [0.04, 0.16]; test-retest reliability by at least 0.19 [0.08, 0.3]; correlation with receptive vocabulary by at least 0.04 [-0.05, 0.12] and correlation with productive vocabulary by at least 0.09 [0.05, 0.13]. Similar patterns hold for different accuracy windows (Figure S6) and across different age groups (Figure S8).

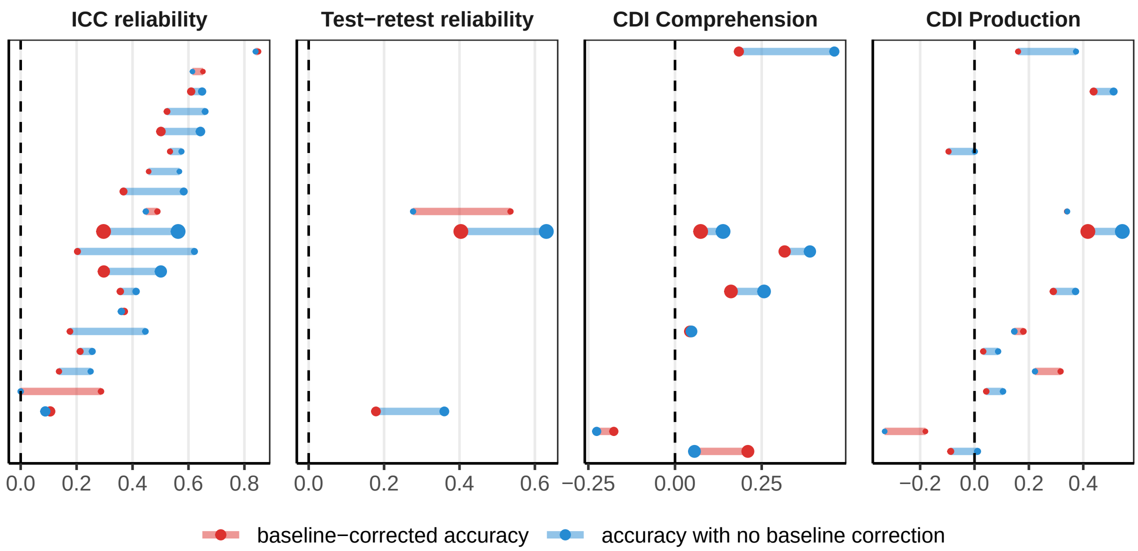

Figure 4. Comparison between a baseline-corrected accuracy in red and accuracy in blue across datasets and measures. Both use an accuracy window of 600 - 4000ms, the baseline window was -2000 - 0ms.

Discussion. Why would incorporating baseline information lead to worse metrics? The literature on the psychometrics of difference scores (e.g., Lord, 1956; Rogosa & Willett, 1983; Trafimow, 2015) provides a valuable guide for evaluating the expected measurement properties of baseline correction. This literature points to two important features of looking while listening tasks: first, children's looking behavior is noisy from trial to trial, meaning that the reliability of proportion looking during both the baseline and the critical window is low; and second, the signal across baseline and critical portions of a trial is typically correlated (in part due to the very salience that the baseline correction is trying to remove). Unfortunately, it is under precisely these conditions – low reliability and high correlation between components – that difference scores exhibit the lowest reliability (Trafimow, 2015). Thus, while baseline-correction may lead to less biased grand average accuracy scores (by removing item salience), it also leads to less reliable and valid individual-level measures.

Recommendation 3: Measure land-based reaction times with a minimum reaction time of around 400 ms.

In addition to accuracy measures, LWL studies also provide the potential for computing children's reaction time [RT; Fernald et al. (2008)], i.e., the speed with which the process familiar words. RT is typically computed on trials when the child is initially looking at the distractor and shifts to the target after stimulus onset, to provide a measure of the speed with which infants respond to the familiar word by shifting their gaze to its target. RT computations include many choice points:

- **Launching vs. landing**: RTs can be computed based on when a child's eyes leave from the distractor (launch-based RT) or when a child's eyes arrive on the target image (land-based RT) (Fernald et al., 2008).

- **Exclusion of short and long RTs**: Short and long RTs are often excluded because they are considered unlikely to be generated in response to the stimulus (Fernald et al., 2008).
- **Linear vs. logarithmic analysis**: Reaction time data are often analyzed using a logarithmic transform (consistent with the idea that reaction times are log-normally distributed and that effects on them may be multiplicative rather than additive) (Csibra, Hernik, Mascaro, Tatone, & Lengyel, 2016; Frank et al., 2025).

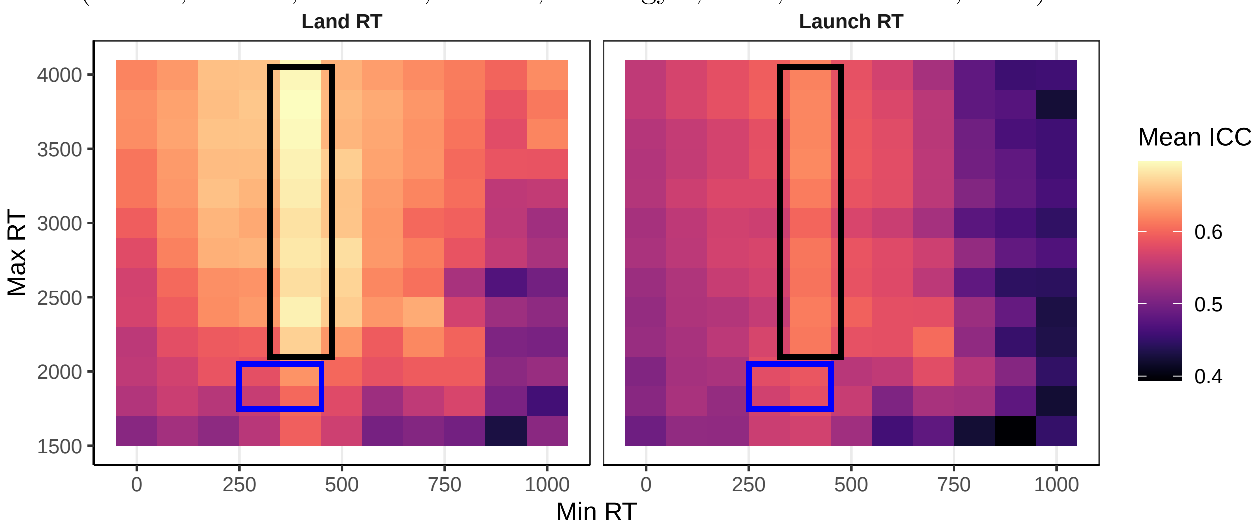

Figure 5. Heat map of different parameter options for measuring RT and how they on average affected ICC.

Analytic approach. We considered a wide range of options for measuring RT, fulling crossing a) launch- or land-based RTs, b) a minimum RT threshold for inclusion between 0 and 1000 ms, c) a maximum RT threshold for inclusion between 1600 and 4000, and d) raw or log-transformed RTs.

Results. An initial exploration of ICC reliability (Figure 5) indicated that ICC reliability was maximized for land-based RTs; a minumum threshold for including RTs set at around 400 ms; and with higher maximum thresholds for RT inclusion. A comparison between a common set of parameters for calculating RT and an alternative that maximizes ICC are shown in Figure 6. Compared to computing the "standard" launch-based, raw RTs restricted to 300-1800ms, using land-based log RTs ranging between 400-4000ms is expected to increase ICC by 0.11 [-0.01, 0.23], with little change in test-retest reliability (0.04 [-0.18, 0.27]). These choices had little impact on CDI correlations (-0.04 [-0.17, 0.08] for comprehension, and -0.03 [-0.12, 0.06] for production). Similar patterns hold for children of different ages (Figure S9 and Figure S11).

Discussion. Overall, RT measures complement accuracy measures and capture substantial signal, as indicated by the moderate ICC values across all choice parameters. A wide range of parameter options have similar properties (Figure S10 and Figure S12), but computing land-based RTs with an inclusion window of 400 - 4000ms maximizes reliability without negatively impacting validity. Note that the recommendation to compute land-based reaction times contradicts traditional approaches to reaction-time computation (Fernald et al., 2008) based on launch time; however, land-based RTs generally led to higher ICCs across

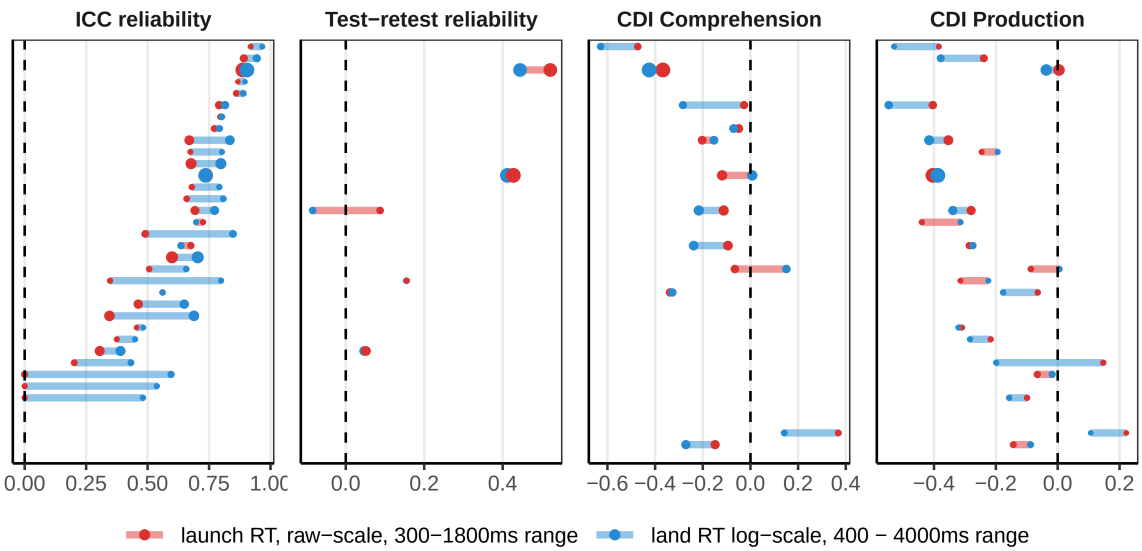

Figure 6. Comparison between two different sets of RT parameters. In red, an example of common practice, using a launch-based RT from 300-1800ms. In blue, an RT that maximizes ICC, with a land RT from 400 to 4000ms, on the log scale. Note that correlations with the CDI are generally expected to be negative (faster word recognition corresponds to more

our dataset. Log-scale transformations should be considered, but may depend on modelling and theoretical considerations.

Recommendation 4: Trial-level and session-level exclusions for low amounts of data are generally unnecessary.

A common issue in developmental research is when to apply trial- and session-level exclusions. For LWL studies, trial-level exclusions tend to be for contributing too few time points (e.g., due to tracker loss or looking away from the screen during the trial). Papers often use a combination of heuristic exclusions for inattention prior to coding looking behavior (Adams et al., 2018) and criterion-based exclusions trials based on the proportion of looking to the target and distractor during the trial. Cutoffs vary, with different experiments requiring minimum looking at the images of at least 20%, 50% or 66% of the critical window of a trial (Borovsky & Peters, 2019; Pomper & Saffran, 2019; Weaver et al., 2024). Studies also occasionally filter for looking at the target or distractor specifically at the point of disambiguation (C. E. Potter, Fourakis, Morin-Lessard, Byers-Heinlein, & Lew-Williams, 2019).

In addition to trial-level exclusions, studies also implement session-level (or child-level) exclusions based on the minimum number of valid trials in a sesson. For accuracy measures, children are often excluded if they had fewer than 25% or 50% of the total trials in a session, either within a condition, or overall (Adams et al., 2018; Garrison, Baudet, Breitfeld, Aberman, & Bergelson, 2020; Perry & Saffran, 2017). For RT, it is instead standard to require a minimum number of trials [e.g., two; Fernald et al. (2006); Adams et al. (2018)].

There are multiple possible ways that trial- and session-level exclusions could affect measurement properties. If trials with fewer valid looking points still provide signal of the child-level language behavior, excluding them will decrease reliability and validity. On the other hand, if these trials reflect noise and are inconsistent with behavior measured on trials with longer looking times, excluding them will increase reliability and validity. Similar possible outcomes apply to session-level exclusions: if children completing only a small number of trials per session are largely contributing estimates with increased noise, reliability and validity should increase when these sessions (or participants) are excluded; conversely, if these children are still contributing consistent signal, reliability and validity should decrease when session-level exclusions are applied.

Analytic approach. We consider 3 types of exclusions: trial-level exclusions for accuracy, administration-level exclusions for accuracy, and administration-level exclusions for RT (an administration in the Peekbank schema corresponds to one session for one child). We first explored trial-level accuracy exclusions, both on the basis of amount of off-target looks and on the basis of where the child was looking at target onset. For accuracy, we then considered per-session minimums either based on the number of valid trials, or based on the fraction of valid trials in an administration. The fraction of valid trials is the more common exclusion in the literature, but when drawing from many datasets that vary in the total number of trials, minimum trial count may represent a more comparable metric. For RT, we considered different cutoffs for the minimum number of valid trials required for inclusion.

We note that datasets contributed to Peekbank vary in whether excluded data was contributed, or whether only cleaned post-exclusion data was contributed. Thus, some datasets may not show effects of exclusions if those trials or administrations that would be excluded are missing from Peekbank.

Results. For exclusions, there are both considerations around reliability and validity, and around how much of the potentially useable data is being discarded. For trial-level exclusions, a requirement to look at the target and distractor at least 50% of the time results in 85.64 % (range: 27.22% - 98.92%) of trials being retained. These exclusions lead to an expected increase in ICC by 0.03 [0, 0.05] and no substantial change in test-retest reliability (-0.02 [-0.11, 0.08]). Differences in CDI correlation are not substantive (0.01 [-0.04, 0.06] for comprehension and 0 [-0.03, 0.03] for production).

Overall, there is a trade-off between reducing the amount of data and increasing the reliability of the remaining data. Filtering based on initial looking position, or having requirements for looking above .5 start to diminish the amount of data, and potentially negatively effect test-retest reliability and CDI measures (Figure S13 and Figure S14). Requirements for on-image looking rates between 0% (i.e., no exclusions) and 50% lead to globally similar outcomes numerically.

We then explored the effects of excluding sessions for too few trials. Here, we focus on the results for a minimum number of 8 total trials, with no trial-level exclusions, to illustrate general patterns in the results, but report outcomes for a range of proportion- and count-based thresholds in the supplement (Figures S15-S18). Only including sessions with at least 8 trials results in retaining 80.99 % (range: 0% - 100%) of total sessions. These exclusions show little evidence of increasing ICC (0 [-0.01, 0.02]) and or test-retest reliability

(0.03 [-0.08, 0.14]) compared to retaining all sessions. Differences in CDI correlation are not substantive (0.01 [-0.04, 0.07] for comprehension and 0.01 [-0.03, 0.05] for production). We do not observe significant differences based on minimum trials below 10 or minimum fractions (Figures S15-S18). Overall, excluding sessions with too few trials had minimal impact on reliability and validity.

We also considered the effects of minimum trial requirements on RT data. Only including sessions that contribute at least 2 trials results in retaining 90.93 % (range: 42.86% - 100%) of all sessions. Similar to previous results, these exclusions lead to a negligible change in ICC by -0.01 [-0.07, 0.04], a slight numerical increase in test-retest reliability by 0.03 [-0.08, 0.14], no change in CDI comprehension correlation(0 [-0.15, 0.14]), and a slight numerical increase in CDI production correlation (-0.02 [-0.08, 0.04]). For RT, setting a low minimum threshold for number of trials (e.g., 2 trials) appears to lead to moderate improvements in test-retest reliability and slightly stronger correlations with CDI production (Figures S19 and S20).

Discussion. Overall, exclusions based on amount of data per-trial or per-session come with trade-offs between the amount of data and variability within the data. Standard practice within the field often errs on the side of excluding a large number of trials and sessions for missing data. Surprisingly, our results show that applying no exclusions or minimal exclusions often represents a reasonable choice, at least in terms of the psychometric properties of the resulting measurements.

Design recommendations

Our focus for this work was to explore how data processing choices affected the reliability and validity of the resultant measures. However, along the way, we identified several experimental design questions that had notable consequences for our central measures. We summarize two design recommendations that stood out as having the largest potential impact on reliability and validity (see supplement, section S8 for an additional analysis of order effects).

Recommendation 5: Design studies to have a minimum of 10 trials per child, per condition, or as many trials as possible for individual-difference studies.

Excluding children for low trial counts may increase test-retest reliability, but it also decreases the power and generalizability of results by excluding some children. Rather than retrospective exclusions, another approach is prospective: how many trials per child should one aim for to achieve adequate reliability and validity? Test-retest reliability is important for studies that investigate longitudinal effects, as it assumes that the test sessions are representative of properties of the child at that age (Byers-Heinlein et al., 2022). To address this prospective question, we considered how reliability and validity measures would have changed if we had fewer trials per child. We filtered the data to include only children who had contributed at least N trials (for different values of N) and then repeatedly subsampled their trials and calculated our metrics on each subset. Datasets vary in how many trials they have, so different minimum trials include different distributions and datasets, items, and children.

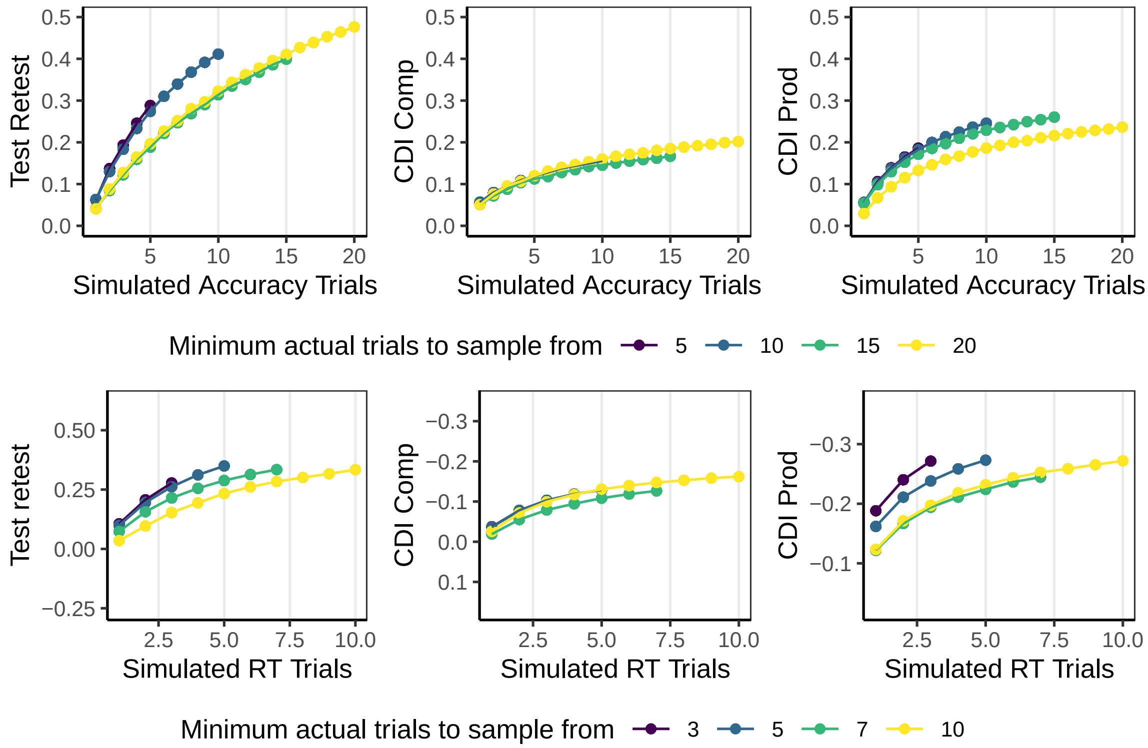

Figure 7. Simulations of measurement properties for different trial counts, created by downsampling trials for all children with different trial minimums. Shown for accuracy window of 600 to 4000ms with no trial- or kid- level exclusions.

The results for accuracy and RT are shown in Figure 7. ICC reliability is not sensitive to the number of trials per subject, so it is not shown. For accuracy, more trials per administration increases test-retest reliability, up to 20 trials (the highest we measured), although the gains per additional trial decrease with more trials. Validity with CDI measures also increased as trials increased, with smaller gains past 5 trials. Based on accuracy, we recommend at least 10 trials per condition in general, and as many as possible for individual differences studies where test-retest reliability is essential.

Similar considerations apply for RT, but at lower total trial numbers. Nearly every trial a child does results in a useable accuracy measure, but only around 1/3 of trials result in a valid RT measure in our data. For test-retest reliability, having at least 5 RT measures per administration leads to notable improvements; gains beyond 5 are hard to measure because of the low number of subjects with sufficient trials, but more trials appears to be beneficial. For CDI measures, there are substantial gains for having 2 trials compared to 1 and continued benefits to including additional trials. Overall, for measuring individual differences in processing speed, designing studies to include as many trials as possible appears to be central to adequate measurement.

Recommendation 6: Match target and distractor images based on animacy.

One concern with LWL trial design is that children may have preferences for which image to look at separate from the auditory stimuli, and some of their looking behavior may be independent of the auditory stimulus. One approach is to decrease the preference discrepancy between a pair of images, so that in the absence of auditory cues, children prefer to look at both images approximately equally. While there are a number of factors that could play into children's image preferences (Fernald et al., 2008; Nothdurft, 2000; Pomper & Saffran, 2019), one factor that stood out in the data was animacy. Children have a strong preference for looking at animals and humans. When trials have one animate and one inanimate, children look more at the animate in the pre-stimulus period (Figure 8A) and are slower to respond when the inanimate is named. In contrast, for trials with target and distractor items matched on animacy, children's early looking patterns for animate and inanimate targets are highly similar. This difference is particularly relevant for the younger children, whose stimulus-contingent looking behavior is weakest, so more of the looking behavior is driven by animacy; however, effects of animacy were apparent in children up to the age of three (Figure 8B).

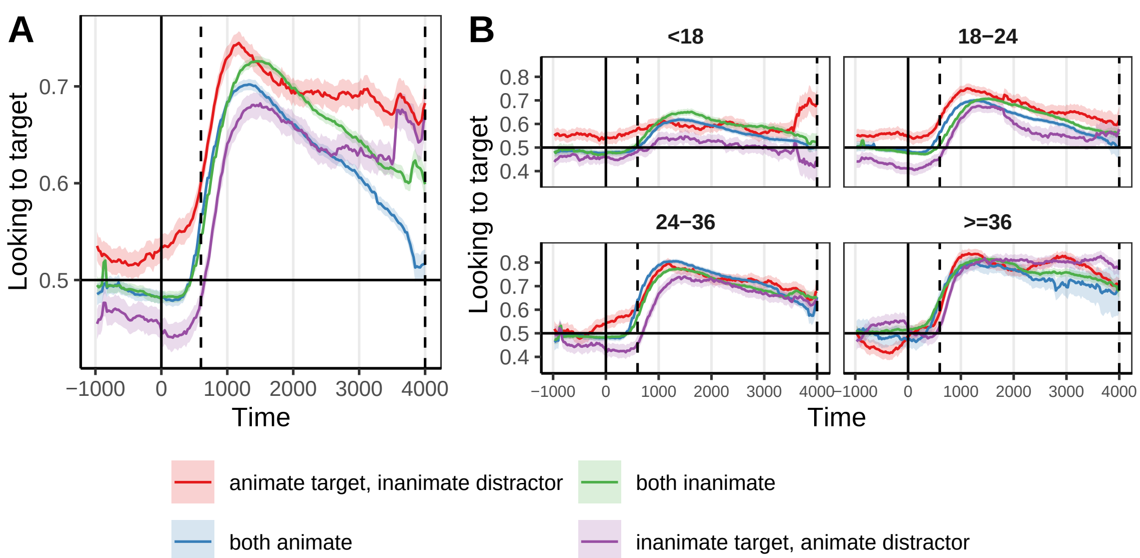

Figure 8. Looking time profiles for different combinations of target and distractor animacy. Panel A shows overall; Panel B shows split by age category.

How much can this matter?

To quantify the potential cumulative consequences of different combinations of choices, we computed the difference in reliability when making all of the recommended choices with respect to data processing to an instance in which a researcher made a set of unfortunate choices that are nevertheless consistent with the literature. Consider two researchers analyzing accuracy in the same LWL dataset, Norm and Rex. Researcher Norm makes a set of reasonable choices based on practices observed in other LWL papers, choosing to analyze a short analysis time window (300-1800 ms); apply baseline correction (baseline window: -2000-0 ms); exclude trials on which infants are not looking for at least 60% of the critical

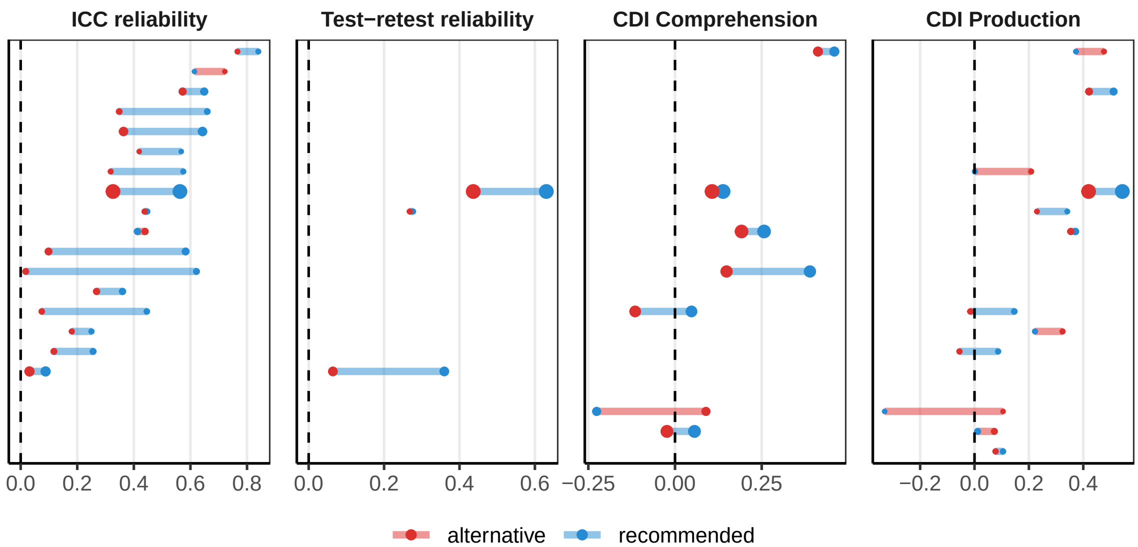

Figure 9. Differences in dependent measures between the recommended accuracy measurement and a sensible-sounding, but unfortunate alternative set of parameters.

window or not looking at the target onset for insufficient attentiveness; and require that infants contribute at least 50% of all trials for inclusion. Researcher Rex, on the other hand, analyzes the same data and instead chooses parameters based on the data-driven recommendations above: a critical window of 600-4000ms, no baseline correction, and including all trials and sessions. Researcher Rex would gain substantial increases in ICC (0.2 [0.13, 0.27]) and test-retest reliability (0.21 [0.13, 0.29]). Additionally, Rex could expect small increases in validity for correlations with comprehension (0.07 [-0.03, 0.17]) and production (0.07 [0, 0.13]). Despite both researchers making choices consistent with past practices, we would expect Rex to derive a measure with significantly improved psychometric properties compared to Norm, and therefore be in a substantially better position to study individual differences and longitudinal change.

General Discussion

Children's early word recognition is a central construct for the study of language learning, meaning that improving recognition measures can have widespread benefits for our understanding of language development. How can we maximize reliable, valid signal in the analysis of data in the LWL paradigm? We took a practical, data-driven approach to this question, using a large-scale dataset of infant LWL data to quantify the impact of different analysis choices on measurement properties, leading to six core recommendations for data processing and experimental design.

For data processing, our strongest take-homes concern window choice and baseline correction. First, longer time windows (e.g., windows that end around 3000 ms post target word onset) maximize the reliability of accuracy measures, with no negative consequences for validity. While average accuracy tends to peak earlier in the timecourse of a trial (typically 1000-2000 ms post target word onset), leading shorter windows to be preferred in the literature, these findings suggest that later looking windows still capture signal that boosts the consistency of word recognition measures. Second, baseline correction, while an intuitive solution to imbalances in the saliency of images, causes large decrements in reliability; we therefore recommend avoiding this practice. Finally, we formulate recommendations for choosing reaction time cutoffs (around 400ms tends to maximize reliability) and find that implementing trial- and session-level exclusions based on the amount of data contributed tend to have minimal consequences. Overall, choices about windows of analysis, approaches to exclusions, and how scores are computed can all have meaningful impacts on measure reliability.

One question that emerges from our analysis concerns the purity of the construct that LWL measures. In our analysis, long accuracy windows increased reliability more than validity. One speculative reason for this difference is that signal from later in the timecourse might be less coupled to language per se, and might instead reflect more general features of attention. Future research could use this difference as a starting point to further parcellate the attentional and linguistic contributors to language-driven eye-movements.

A more general take-home from our data processing analyses is that intuitive data-based practices, e.g., looking at patterns in grand-average time courses, may not always be a trustworthy guide to child- or item-level psychometric properties. Past recommendations for time window selection have sometimes been motivated by inspecting consistent patterns of looking in grand-averages and noting when looking begins to decline across participants. These heuristics — while based in important patterns in looking behavior — may not always lead to optimal choices with respect to core measurement properties such as reliability. Wherever possible, we recommend that measurement choices be guided by direct estimates of psychometric properties in order to maximize the signal in infant looking behavior (Byers-Heinlein et al., 2022; Sander-Montant et al., 2025; Weaver et al., 2026).

Among our design-focused recommendations, the most striking result is the impact of animacy on infant looking behavior. When an image of an animate referent was paired with an image of an inanimate referent, infant looking was systematically biased towards the animate image. This finding has important consequences for estimating word recognition: accuracy for animate targets is substantially overestimated and accuracy for inanimate targets becomes substantially underestimated when animacy is not balanced in the target-distractor pairing. The results are especially dramatic for younger infants (under 18 months): looking for familiar inanimate targets can appear to be at chance when paired with an animate distractor. In other words, an improper balancing of saliency could make an infant appear to not know a word, even though they do. These results emphasize the importance of balancing saliency in image pairings as much as possible in LWL designs. This is especially true given that many of the current analytic approaches to correcting for saliency imbalances post hoc rely on difference-based measures (baseline correction; item-pair correction) which decrease reliability.

Limitations & Generalizability

The scope of our recommendations is fundamentally limited by the characteristics of the Peekbank dataset. First, while the Peekbank database is by far the largest dataset of its kind, it is not a comprehensive or representative set of LWL studies. To highlight just a few of these limitations, (1) the data is predominantly English word recognition collected from infants living in the U.S., and (2) the dataset is a convenience sample from labs who made their data openly available or otherwise accessible on request from the Peekbank team. LWL studies as a whole are likely more heterogeneous than what is represented in Peekbank — the datasets considered here come predominantly from a small group of labs or networks of researchers with similar training in LWL data collection.

At the same time, there is also substantial heterogeneity in the studies included in Peekbank, due to many studies pursuing different research questions and therefore making a variety of distinct design choices. To mitigate this concern, our analyses focused exclusively on what we term "vanilla" trials, i.e., trials that present familiar words with familiar images with limited or no targeted manipulations of the stimuli. The main benefit of focusing on vanilla trials is that it allows for a clear interpretation of reliability and validity — trials are truly designed to measure word recognition, rather than some other aspect of language processing. We view the remaining heterogeneity of study properties (e.g., specific words; differences in age) as a strength of our approach — our findings generalize broadly across a wide range of items and ages. However, by focusing on vanilla trials only, we also limit our conclusions to measuring familiar word recognition; our findings do not address how data processing and design practices would generalize to other estimation goals, e.g., estimating the magnitude of differences between two experimental conditions (though there is good reason to think that some of the more consequential choices, e.g. baseline correction, are likely have a similar impact).

While we summarize the choices that typically maximize reliability, we note that we often see a range of choices with similar outcomes in our analyses. For example, in determining the best time window for computing accuracy, a choice of ending a time window at 2800 vs. 3000 or 3000 ms vs. 3200 ms leads to very similar outcomes. There is no one choice of time window that we can definitively infer as the "best" for maximizing reliability — instead, we see a pattern of graded improvements in reliability that follow a predictable pattern. For example, longer time windows tend to increase reliability holding the onset of the accuracy window constant. While we sometimes articulate which choices lead to the highest reliability descriptively, these are at best approximations. We offer these windows as data-driven default choices (especially in cases where the decision would otherwise be ad hoc).

Conclusions

Experimental design and data processing decisions can have profound impacts on the properties of infant measures. Our findings show how compiling large-scale datasets such as Peekbank can help us estimate the impact of a wide variety of experimenter decisions, generalizable across a wide range of ages, stimuli, and experimental contexts. Establishing consistent practices and principles for reliable and valid measurement of infant looking sets the stage for asking central questions about how language skill develops.

References

Adams, K. A., Marchman, V. A., Loi, E. C., Ashland, M. D., Fernald, A., & Feldman, H. M. (2018). Caregiver talk and medical risk as predictors of language outcomes in full term and preterm toddlers. *Child Development*, *89*(5), 1674–1690.

Bacon, D., & Saffran, J. (2022). Role of speaker gender in toddler lexical processing. *Infancy : The Official Journal of the International Society on Infant Studies*, *27*(2), 291–300.

Bergelson, E., & Aslin, R. (2017). Semantic Specificity in One-Year-Olds' Word Comprehension. *Language Learning and Development*, *13*(4), 481–501. https://doi.org/10.1080/15475441.2017.1324308

Bergelson, E., & Swingley, D. (2012). At 6-9 months, human infants know the meanings of many common nouns. *Proceedings of the National Academy of Sciences*, *109*(9), 3253–3258.

Bergmann, C., Tsuji, S., Piccinini, P. E., Lewis, M. L., Braginsky, M., Frank, M. C., & Cristia, A. (2018). Promoting replicability in developmental research through meta-analyses: Insights from language acquisition research. *Child Development*, *89*(6), 1996–2009.

Borovsky, A., & Peters, R. E. (2019). Vocabulary size and structure affects real-time lexical recognition in 18-month-olds. *PLOS ONE*, *14*(7), e0219290. https://doi.org/10.1371/journal.pone.0219290

Byers-Heinlein, K., Bergmann, C., & Savalei, V. (2022). Six solutions for more reliable infant research. *Infant and Child Development*, *31*(5), e2296. https://doi.org/10.1002/icd.2296

Csibra, G., Hernik, M., Mascaro, O., Tatone, D., & Lengyel, M. (2016). Statistical treatment of looking-time data. *Developmental Psychology*, *52*(4), 521–536.

Fernald, A., & Marchman, V. A. (2012). Individual differences in lexical processing at 18 months predict vocabulary growth in typically developing and late-talking toddlers. *Child Development*, *83*(1), 203–222.

Fernald, A., Perfors, A., & Marchman, V. A. (2006). Picking up speed in understanding: Speech processing efficiency and vocabulary growth across the 2nd year. *Developmental Psychology*, *42*(1), 98–116. https://doi.org/10.1037/0012-1649.42.1.98

Fernald, A., Pinto, J. P., Swingley, D., Weinberg, A., & McRoberts, G. W. (1998). Rapid gains in speed of verbal processing by infants in the 2nd year. *Psychological Science*, *9*(3), 228–231.

Fernald, A., Zangl, R., Portillo, A. L., & Marchman, V. A. (2008). Looking while listening: Using eye movements to monitor spoken language comprehension by infants and young children. In I. A. Sekerina, E. M. Fernández, & H. Clahsen (Eds.), *Language Acquisition and Language Disorders* (Vol. 44, pp. 97–135). Amsterdam: John Benjamins Publishing Company. https://doi.org/10.1075/lald.44.06fer

Flake, J. K., & Fried, E. I. (2020). Measurement Schmeasurement: Questionable Measurement Practices and How to Avoid Them. *Advances in Methods and Practices in Psychological Science*, *3*(4), 456–465. https://doi.org/10.1177/2515245920952393

Frank, M. C., Braginsky, M., Yurovsky, D., & Marchman, V. A. (2017). Wordbank: An open repository for developmental vocabulary data. *Journal of Child Language*, *44*(3), 677–694.

Frank, M. C., Braginsky, M., Yurovsky, D., & Marchman, V. A. (2021). *Variability and* *consistency in early language learning: The wordbank project*. Cambridge, MA: MIT Press.

Frank, M. C., Marchman, V. A., Bergey, C. A., Boyce, V., Braginsky, M., Kachergis, G., … Zettersten, M. (2025). Continuous developmental changes in word recognition support language learning across early childhood. *eLife*. https://doi.org/10.7554/eLife.109636.1

Garrison, H., Baudet, G., Breitfeld, E., Aberman, A., & Bergelson, E. (2020). Familiarity plays a small role in noun comprehension at 12–18 months. *Infancy : The Official Journal of the International Society on Infant Studies*, *25*(4), 458–477.

Golinkoff, R. M., Ma, W., Song, L., & Hirsh-Pasek, K. (2013). Twenty-five years using the intermodal preferential looking paradigm to study language acquisition: What have we learned? *Perspectives on Psychological Science*, *8*(3), 316–339.

Hirsh-Pasek, K., & Golinkoff, R. M. (1996). The intermodal preferential looking paradigm: A window onto emerging language comprehension. In *Language, Speech, and Communication. Methods for assessing children's syntax* (pp. 105–124). Cambridge, MA, US: The MIT Press.

Hurtado, N., Marchman, V. A., & Fernald, A. (2007). Spoken word recognition by Latino children learning Spanish as their first language. *Journal of Child Language*, *34*(2), 227–249. https://doi.org/10.1017/S0305000906007896

Kominsky, J. F., Lucca, K., Thomas, A. J., Frank, M. C., & Hamlin, J. K. (2022). Simplicity and validity in infant research. *Cognitive Development*, *63*, 101213. https://doi.org/10.1016/j.cogdev.2022.101213

Law, F., II, & Edwards, J. R. (2015). Effects of Vocabulary Size on Online Lexical Processing by Preschoolers. *Language Learning and Development*, *11*(4), 331–355. https://doi.org/10.1080/15475441.2014.961066

Loi, E. C., Marchman, V. A., Fernald, A., & Feldman, H. M. (2017). Using eye movements to assess language comprehension in toddlers born preterm and full term. *The Journal of Pediatrics*, *180*, 124–129. https://doi.org/10.1016/j.jpeds.2016.10.004

López Pérez, M., Moore, C., Sander-Montant, A., & Byers-Heinlein, K. (2024). Infants' Knowledge of Individual Words: Investigating Links Between Parent Report and Looking Time. *Infancy*, *30*(1), e12641. https://doi.org/10.1111/infa.12641

Lord, F. M. (1956). The Measurement of Growth. *Educational and Psychological Measurement*, *16*(4), 421–437. https://doi.org/10.1177/001316445601600401

Mahr, T., McMillan, B. T. M., Saffran, J. R., Ellis Weismer, S., & Edwards, J. (2015). Anticipatory coarticulation facilitates word recognition in toddlers. *Cognition*, *142*, 345–350. https://doi.org/10.1016/j.cognition.2015.05.009

Marchman, V. A., Adams, K. A., Loi, E. C., Fernald, A., & Feldman, H. M. (2016). Early language processing efficiency predicts later receptive vocabulary outcomes in children born preterm. *Child Neuropsychology*, *22*(6), 649–665. https://doi.org/10.1080/09297049.2015.1038987

Marchman, V. A., Dale, P., & Fenson, L. (2023). *MacArthur-bates communicative development inventories user's guide and technical manual*. Brookes Publishing.

Marchman, V. A., Loi, E. C., Adams, K. A., Ashland, M., Fernald, A., & Feldman, H. M. (2018). Speed of language comprehension at 18 months old predicts school-relevant outcomes at 54 months old in children born preterm. *Journal of Developmental & Behavioral Pediatrics*, *39*(3), 246–253.

Mirman, D., Dixon, J. A., & Magnuson, J. S. (2008). Statistical and computational models of the visual world paradigm: Growth curves and individual differences. *Journal of Memory and Language*, *59*(4), 475–494. https://doi.org/10.1016/j.jml.2007.11.006

Moore, C., & Bergelson, E. (2024). Wordform variability in infants' language environment and its effects on early word learning. *Cognition*, *245*, 105694. https://doi.org/10.1016/j.cognition.2023.105694

Nothdurft, H.-C. (2000). Salience from feature contrast: Additivity across dimensions. *Vision Research*, *40*(10-12), 1183–1201.

Novack, M. A., Dworak, E. M., Han, Y. C., Kaat, A. J., Ustsinovich, V., Saffran, J., ... Gershon, R. C. (2025). Development and validation of the NIH Baby Toolbox® language measures. *Infant Behavior and Development*, *80*, 102121. https://doi.org/10.1016/j.infbeh.2025.102121

Olson, R. H., Pomper, R., Potter, C. E., Hay, J. F., Saffran, J. R., Ellis Weismer, S., & Lew-Williams, C. (2020). *Peyecoder: An open-source program for coding eye movements.* https://doi.org/10.5281/zenodo.3939234

Peelle, J. E., & Van Engen, K. J. (2021). Time Stand Still: Effects of Temporal Window Selection on Eye Tracking Analysis. *Collabra: Psychology*, *7*(1), 25961. https://doi.org/10.1525/collabra.25961

Perry, L. K., & Saffran, J. R. (2017). Is a pink cow still a cow? Individual differences in toddlers' vocabulary knowledge and lexical representations. *Cognitive Science*, *41*(4), 1090–1105. https://doi.org/10.1111/cogs.12370

Peter, M. S., Durrant, S., Jessop, A., Bidgood, A., Pine, J. M., & Rowland, C. F. (2019). Does speed of processing or vocabulary size predict later language growth in toddlers? *Cognitive Psychology*, *115*, 101238.

Pomper, R., & Saffran, J. R. (2016). Roses are red, socks are blue: Switching dimensions disrupts young children's language comprehension. *PloS One*, *11*(6), e0158459. https://doi.org/10.1371/journal.pone.0158459

Pomper, R., & Saffran, J. R. (2019). Familiar object salience affects novel word learning. *Child Development*, *90*(2), e246–e262. https://doi.org/10.1111/cdev.13053

Potter, C. E., Fourakis, E., Morin-Lessard, E., Byers-Heinlein, K., & Lew-Williams, C. (2019). Bilingual toddlers' comprehension of mixed sentences is asymmetrical across their two languages. *Developmental Science*, *22*(4), e12794. https://doi.org/10.1111/desc.12794

Potter, C., & Lew-Williams, C. (unpublished). *Behold the canine!: How does toddlers' knowledge of typical frames and familiar words interact to influence their sentence processing?*

Reznick, J. S. (1990). Visual preference as a test of infant word comprehension. *Applied Psycholinguistics*, *11*(2), 145–166.

Rhemtulla, M., & Bottesini, J. G. (2022). On the reliability of reliability estimates in infancy research. *Infant and Child Development*, *31*(5), e2327. https://doi.org/10.1002/icd.2327

Rogosa, D. R., & Willett, J. B. (1983). Demonstrating the Reliability the Difference Score in the Measurement of Change. *Journal of Educational Measurement*, *20*(4), 335–343. https://doi.org/10.1111/j.1745-3984.1983.tb00211.x

Ronfard, S., Wei, R., & Rowe, M. L. (2022). Exploring the linguistic, cognitive, and social skills underlying lexical processing efficiency as measured by the looking-while-listening paradigm. *Journal of Child Language*, *49*(2), 302–325. https://doi.org/10.1017/S0305000921000106

Sander-Montant, A., Fibla, L., Byers-Heinlein, K., & Killam, H. (2025). *Keeping an eye on looking measures: Towards more robust developmental methods*. OSF. https://doi.org/10.31234/osf.io/vxu53_v1

Sander-Montant, A., Pérez, M. L., & Byers-Heinlein, K. (2023). The more they hear the more they learn? Using data from bilinguals to test models of early lexical development. *Cognition*, *238*, 105525. https://doi.org/10.1016/j.cognition.2023.105525

Schreiner, M. S., Zettersten, M., Bergmann, C., Frank, M. C., Fritzsche, T., Gonzalez-Gomez, N., et al.others. (2024). Limited evidence of test-retest reliability in infant-directed speech preference in a large preregistered infant experiment. *Developmental Science*, *27*(6), e13551. https://doi.org/10.1111/desc.13551

Snedeker, J. (2013). Children's sentence processing. In *Sentence Processing*. Psychology Press.

Swingley, D., & Aslin, R. N. (2000). Spoken word recognition and lexical representation in very young children. *Cognition*, *76*(2), 147–166. https://doi.org/10.1016/s0010-0277(00)00081-0

Swingley, D., & Aslin, R. N. (2002). Lexical Neighborhoods and the Word-Form Representations of 14-Month-Olds. *Psychological Science*, *13*(5), 480–484. https://doi.org/10.1111/1467-9280.00485

Trafimow, D. (2015). A defense against the alleged unreliability of difference scores. *Cogent Mathematics*, *2*(1), 1064626. https://doi.org/10.1080/23311835.2015.1064626

Venker, C. E., Pomper, R., Mahr, T., Edwards, J., Saffran, J., & Ellis Weismer, S. (2020). Comparing automatic eye tracking and manual gaze coding methods in young children with autism spectrum disorder. *Autism Research*, *13*(2), 271–283. https://doi.org/https://doi.org/10.1002/aur.2225

Von Holzen, K., & Mani, N. (2012). Language nonselective lexical access in bilingual toddlers. *Journal of Experimental Child Psychology*, *113*(4), 569–586. https://doi.org/10.1016/j.jecp.2012.08.001

Wass, S. V., Smith, T. J., & Johnson, M. H. (2013). Parsing eye-tracking data of variable quality to provide accurate fixation duration estimates in infants and adults. *Behavior Research Methods*, *45*, 229–250.

Weaver, H., & Saffran, J. R. (2026). Interrogating Early Word Knowledge: Factors That Influence the Alignment Between Caregiver-Report and Experimental Measures. *Developmental Science*, *29*(1), e70088. https://doi.org/10.1111/desc.70088

Weaver, H., Saffran, J., & Arunachalam, S. (2026). Looking forward: Eye-gaze methods in vocabulary development research. *Trends in Cognitive Sciences*.

Weaver, H., Zettersten, M., & Saffran, J. R. (2024). Becoming word meaning experts: Infants' processing of familiar words in the context of typical and atypical exemplars. *Child Development*, *95*(5), e352–e372. https://doi.org/10.1111/cdev.14120

Weisleder, A., & Fernald, A. (2013). Talking to children matters: Early language experience strengthens processing and builds vocabulary. *Psychological Science*, *24*(11), 2143–2152. https://doi.org/10.1177/0956797613488145

Yurovsky, D., Wade, A., Kraus, A., Gengoux, G., Antonio, Y., & Frank, M. (2017). *Developmental changes in the speed of social attention in early word learning.*

unpublished manuscript.
Zettersten, M., Yurovsky, D., Xu, T. L., Uner, S., Tsui, A. S. M., Schneider, R. M., ... Frank, M. C. (2022). Peekbank: An open, large-scale repository for developmental eye-tracking data of children's word recognition. *Behavior Research Methods*, *55*(5), 2485–2500. https://doi.org/10.3758/s13428-022-01906-4